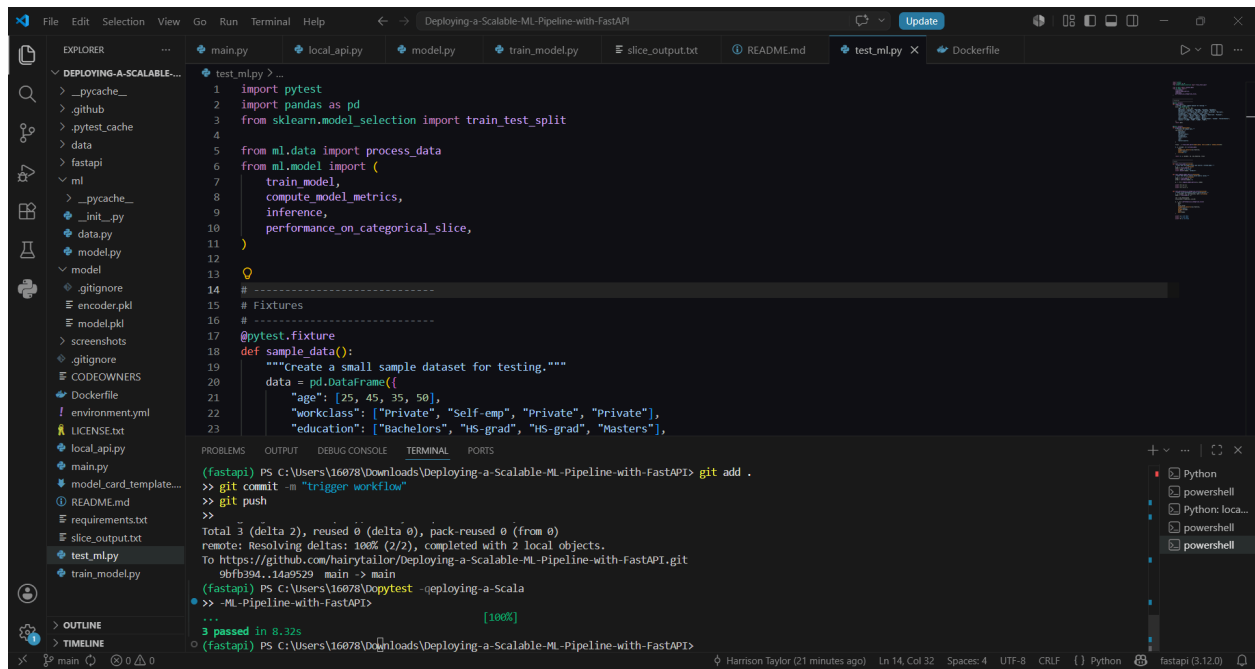
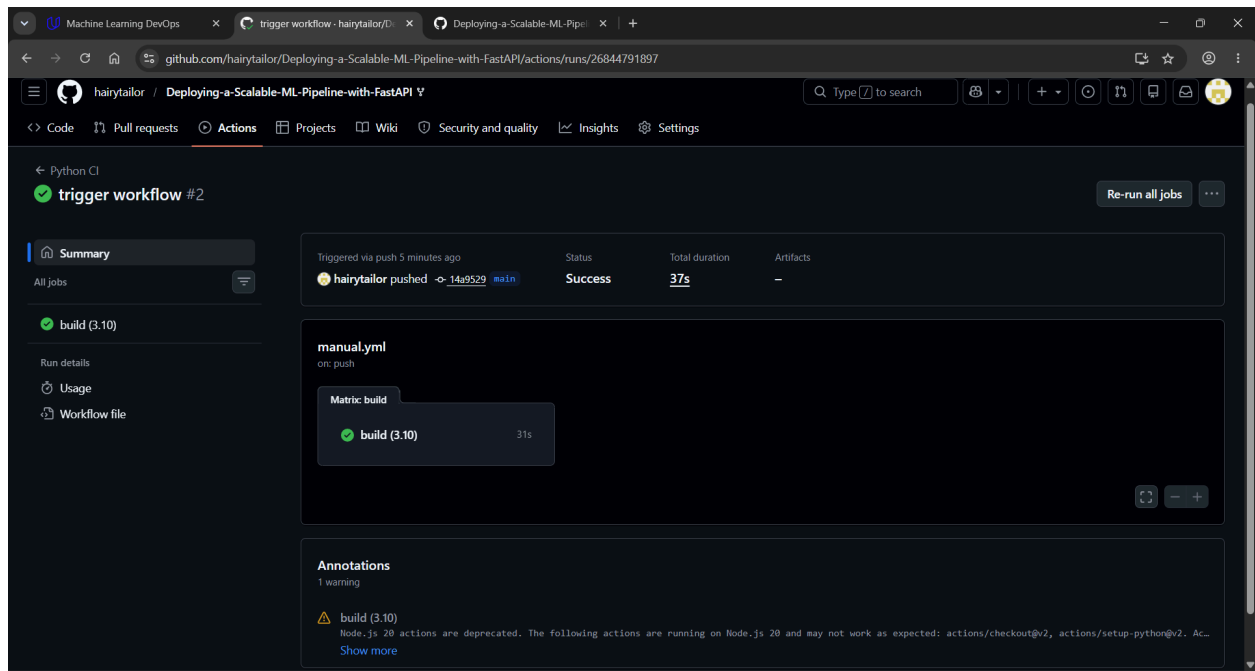
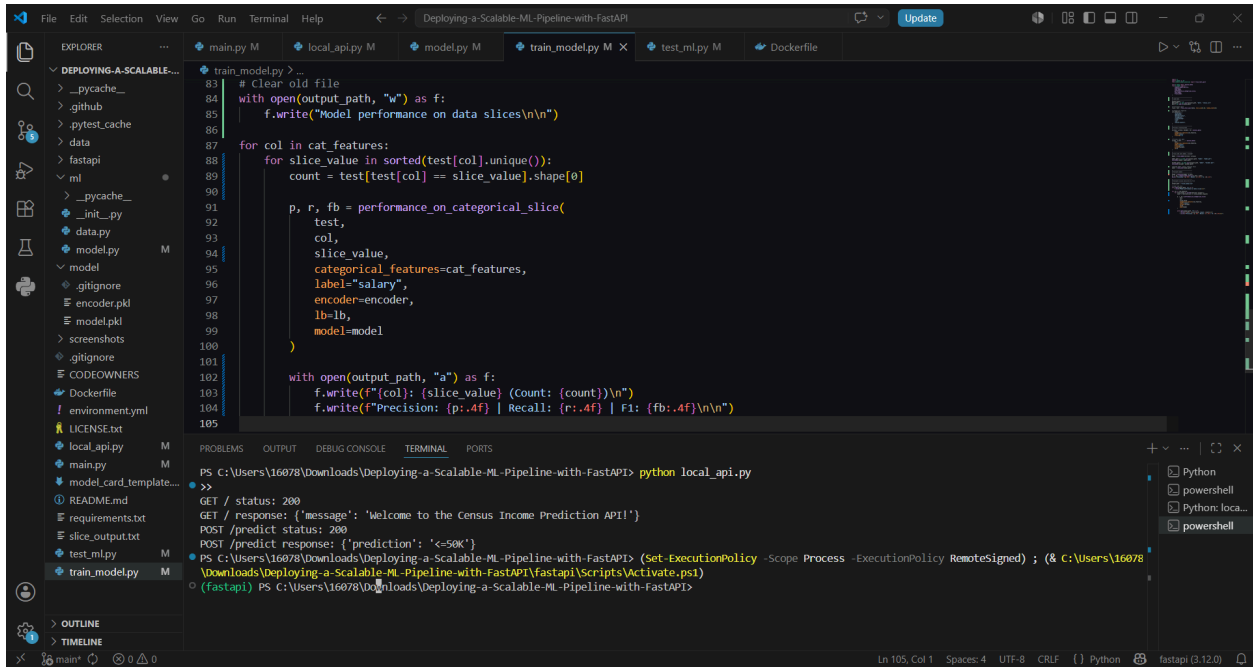


Harrison Taylor

D501 ML

Github Link: <https://github.com/hairytailor/Deploying-a-Scalable-ML-Pipeline-with-FastAPI.git>





```
1 # Model Card
2
3 For additional information see the Model Card paper: https://arxiv.org/pdf/1810.03993.pdf
4
5 ## Model Details
6 This model is a supervised machine learning classifier trained to predict whether an individual earns more than $50,000 per year using the UCI Adult Census dataset. The model is implemented as a Random Forest Classifier and uses one-hot encoding for categorical features and label binarization for the target variable. The preprocessing artifacts (encoder and label binarizer) and the trained model are saved for reproducibility and deployment.
7
8 ## Intended Use
9 The model is intended for educational purposes as part of demonstrating an end-to-end MLOps workflow, including data processing, model training, evaluation, CI/CD, and API deployment. It is not intended for real-world decision-making in hiring, credit scoring, or financial assessment. Any real-world use would require extensive fairness testing, bias mitigation, and regulatory review.
10
11 ## Training Data
12 The model was trained on the Adult Census dataset, which includes demographic and employment-related features such as age, workclass, education, marital status, occupation, relationship, race, sex, and native country. The dataset was split into training and testing sets using an 80/20 split. Categorical features were processed using one-hot encoding, and the target variable was binarized.
13
14 ## Evaluation Data
15 The evaluation data consists of the held-out test split from the same dataset. The same preprocessing pipeline used during training was applied to the evaluation data. Model performance was assessed using standard classification metrics.
16
17 ## Metrics
18 The model was evaluated using precision, recall, and F1 score:
19
20 Precision: 0.7419
21
22 Recall: 0.6384
23
24 F1 Score: 0.6863
25
26 Slice-based performance was also computed across categorical features. Performance varies across groups, with some slices (e.g., "Doctorate", "Prof-school", "Exec-managerial") showing strong performance, while others with small sample sizes or imbalanced labels (e.g., "7th-8th grade", "Greece", "Jamaica") show poor or unstable metrics.
27
```

```
1 workclass: ?, Count: 389
2 Precision: 0.6538 | Recall: 0.4048 | F1: 0.5000
3 workclass: Federal-gov, Count: 191
4 Precision: 0.7971 | Recall: 0.7857 | F1: 0.7914
5 workclass: Local-gov, Count: 387
6 Precision: 0.7576 | Recall: 0.6818 | F1: 0.7177
7 workclass: Private, Count: 4,578
8 Precision: 0.7376 | Recall: 0.6404 | F1: 0.6856
9 workclass: Self-emp-inc, Count: 212
10 Precision: 0.7887 | Recall: 0.7542 | F1: 0.7672
11 workclass: Self-emp-not-inc, Count: 498
12 Precision: 0.7064 | Recall: 0.4904 | F1: 0.5789
13 workclass: State-gov, Count: 254
14 Precision: 0.7424 | Recall: 0.6712 | F1: 0.7050
15 workclass: Without-pay, Count: 4
16 Precision: 1.0000 | Recall: 1.0000 | F1: 1.0000
17 education: 10th, Count: 183
18 Precision: 0.4000 | Recall: 0.1667 | F1: 0.2353
19 education: 11th, Count: 225
20 Precision: 1.0000 | Recall: 0.2727 | F1: 0.4286
21 education: 12th, Count: 98
22 Precision: 1.0000 | Recall: 0.4000 | F1: 0.5714
23 education: 1st-4th, Count: 23
24 Precision: 1.0000 | Recall: 1.0000 | F1: 1.0000
25 education: 5th-6th, Count: 62
26 Precision: 1.0000 | Recall: 0.5000 | F1: 0.6667
27 education: 7th-8th, Count: 141
28 Precision: 0.0000 | Recall: 0.0000 | F1: 0.0000
29 education: 9th, Count: 115
30 Precision: 1.0000 | Recall: 0.3333 | F1: 0.5000
31 education: Assoc-acdm, Count: 198
32 Precision: 0.7000 | Recall: 0.5957 | F1: 0.6437
33 education: Assoc-voc, Count: 273
34 Precision: 0.6471 | Recall: 0.5238 | F1: 0.5789
35 education: Bachelors, Count: 1,053
36 Precision: 0.7523 | Recall: 0.7289 | F1: 0.7404
37 education: Doctorate, Count: 77
```